

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 11

DUE: WEDNESDAY, APRIL 22, 2026, BEFORE 23:59PM

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let $n \in \mathbb{N}$ and let $X_n = (\{0, \dots, n\} \times \Delta^2) / \sim$, where the equivalence relation is generated by $(i, \delta^2(t)) \sim (i, \delta^0(t))$ for all $i \in \{0, \dots, n\}$ and all $t \in \Delta^1$, and $(i, \delta^1(t)) \sim (i-1, \delta^0(t))$ for all $i \in \{1, \dots, n\}$ and all $t \in \Delta^1$.
 - (a) Let $\sigma_i \in \text{Sin}_2(X_n)$ be defined as $\sigma_i(t) = [(i, t)]$ for $i \in \{0, \dots, n\}$. Prove that $d_0(\sigma_i) = d_2(\sigma_i)$ for all $i \in \{0, \dots, n\}$ and $d_1(\sigma_i) = d_0(\sigma_{i-1})$ for all $i \in \{1, \dots, n\}$.
 - (b) Let $a = d_1(\sigma_0)$ and $b_i = d_0(\sigma_i)$ for $i \in \{0, \dots, n\}$. Explain why there is a Δ -complex structure on X_n with $\Sigma_2 = \{\sigma_0, \dots, \sigma_n\}$ and $\Sigma_1 = \{a, b_0, \dots, b_n\}$.

You do not need to rigorously verify the axioms for a Δ -complex. A convincing picture, for instance as “triangles with identifications on their boundaries”, as well as a definition of Σ_0 , is sufficient.
2. *This question is a continuation of the previous.*
 - (a) Prove that $\{b_n, b_{n-1} - 2b_n, \dots, b_0 - 2b_1, a - 2b_0\}$ is a basis for $\Delta_1(X_n)$ and deduce that $H_1(X_n) \cong \mathbb{Z}$. Your proof should identify a generator of $H_1(X_n)$.

Hint: an upper-triangular square matrix with 1's on the diagonal is invertible.
 - (b) The inclusion map $i : X_n \rightarrow X_{n+1}$ determines a homomorphism $H_1(i) : H_1(X_n) \rightarrow H_1(X_{n+1})$. Question 2(a) identifies $H_1(i)$ with a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$. Determine which homomorphism.

3. (a) Sketch a picture of the Klein bottle, as a “square with identifications on its boundary”.
- (b) Sketch why the Klein bottle admits a Δ -complex structure two 2-simplices, three 1-simplices, and one 0-simplex.
You do not need to make precise definitions nor verify the axioms, a convincing picture is sufficient. Your picture should give names to all simplices, for instance $\Sigma_2 = \{\rho, \rho'\}$ and $\Sigma_1 = \{a, b, c\}$, and should be detailed enough to answer question (c) below.
- (c) Determine formulas for all face maps $\Sigma_2 \rightarrow \Sigma_1$ and deduce that $H_2(K) \cong 0$.
4. Let X be a Δ complex, and assume that X is compact. Prove that the Δ -complex structure consists of finitely many simplices.
5. Let $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{Z}_{>0}\}$ have the subspace topology from $\mathbb{R}$. Prove that X does not admit a Δ -complex structure.